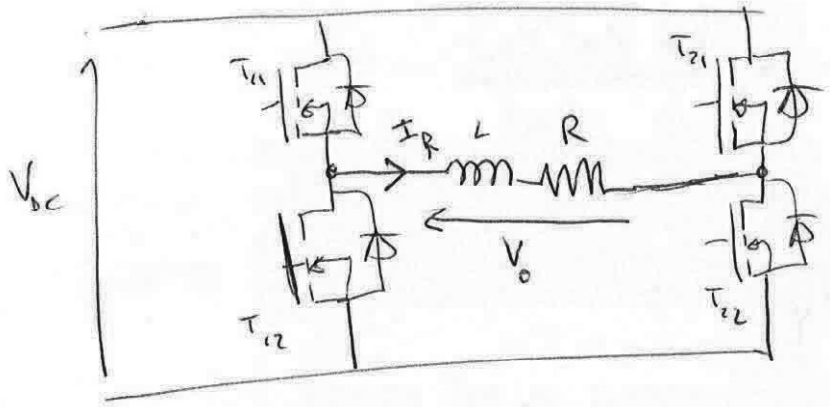
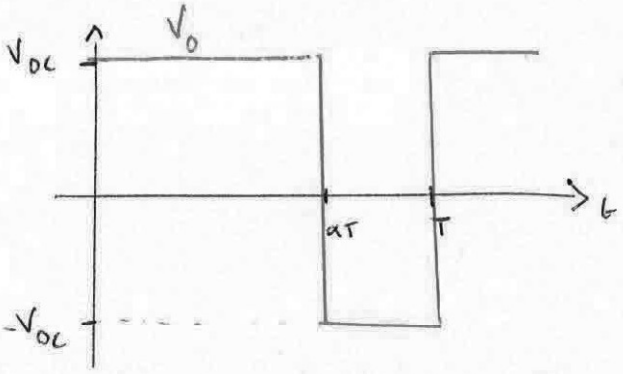
Single-Phase PWM Inverter with RL Load



$\alpha(t) = 0{,}5 + \delta_\alpha \sin(\omega t)$

1.1)



1.2) $\langle V_o \rangle = (2\alpha - 1) V_{DC}$

1.3) $\langle V_o \rangle (t) = (2\alpha(t) - 1) V_{DC}$

$= 2\delta_a V_{DC} \sin \omega t$

1.4) $V_{o\,eff} = \sqrt{2}\, \delta_a V_{DC}$.

1.5) $I_R(t) = I_{R_0} \sin(\omega t - \varphi)$

1.6) $I_{R\,eff} = \dfrac{V_{o\,eff}}{\sqrt{R^2 + (L\omega)^2}}$

1.7) $\tan \varphi = \dfrac{L\omega}{R} \Rightarrow \varphi = \operatorname{atan}\left(\dfrac{L\omega}{R}\right)$

1.8) $pf = \cos\varphi = \cos\left(\operatorname{atan}\left(\frac{L\omega}{R}\right)\right)$

1.9) $P_R = R\, I_{R\,eff}^2 = R\,\frac{V_{0\,eff}^2}{R^2 + L\omega^2} = \frac{2R\,\delta_\alpha^2 V_{DC}^2}{R^2 + (L\omega)^2}$

$= \frac{2\,\delta_\alpha^2 V_{DC}^2}{R} \cdot \frac{1}{1 + \tan^2\varphi}$

2.1) $P_R = 500W$ $\quad U_{R_n} = 100V \Rightarrow I_{R_n} = \frac{P_R}{U_{R_n}} = 5A.$

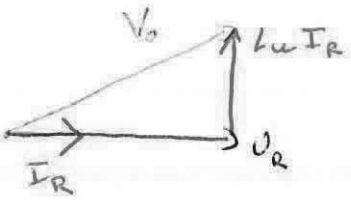
$P_R = R\, I_{R_n}^2 \Rightarrow R = \frac{P_R}{I_{R_n}^2} = 20\Omega$

or : $P_R = 500W = \frac{U_{R_n}^2}{R} \Rightarrow R = \frac{U_{R_n}^2}{P_R} = \frac{100^2}{500} = 20\Omega.$

2.2) $L = 30\,mH$ $\quad f = 100Hz \rightarrow \varphi = \operatorname{atan}\left(\frac{L\omega}{R}\right) = 0{,}75\,rad\ (43°)$

$\Rightarrow \cos\varphi = 0{,}72.$

2.3)



$\underline{V_0} = \underline{U_R} + jL\omega\,\underline{I_R}$

2.4) $I_R = 5A \rightarrow V_L = L\omega I_R = 94{,}2\,V.$

2.5) $V_{0\,eff} = \sqrt{U_R^2 + V_L^2} = \sqrt{100^2 + 94{,}2^2} = 137{,}4V.$

2.6) $V_{0\,eff} = \sqrt{2}\,\delta_\alpha V_{DC}$ with $V_{DC} = 230\sqrt{2} = 325V.$

$\Rightarrow \delta_\alpha = \frac{V_{0\,eff}}{\sqrt{2}\,V_{DC}} = 0{,}30$

2.7) $V_{Res\,min} = 230 - 15\% = 196V \rightarrow \delta_{\alpha\,max} = 0,35$

$V_{Res\,max} = 230 + 15\% = 265V \rightarrow \delta_{\alpha\,min} = 0,26$

2.8) Pour $V_{Res} = 230V$ We have P_{max} for $\delta_\alpha = 0,5$ $(0<\alpha<1)$

$\rightarrow V_{eff\,max} = 230V$ $Z = \sqrt{R^2 + (L\omega)^2} = 27,5\,\Omega$

$\hookrightarrow I_{Reff\,max} = 8,4A$

$\Rightarrow P_{max} = R \cdot I_{Reff\,max}^2 = 1,4\,kW.$

3.) Max voltage : $65\sqrt{2} = 374V$

Max current : $\frac{V_{eff}\sqrt{2}}{\sqrt{R^2+(L\omega)^2}} = \frac{\sqrt{2}\,\delta_\alpha V_{DC}\cdot\sqrt{2}}{\sqrt{R^2+(L\omega)^2}} = 5,7A$